PHOENIX Project — Cybersecurity Threat Analysis

CY004: Identify Entry Points

Understanding who might attack Phoenix is only the first step in building effective defences. The second is understanding exactly where they would attack. An entry point is not merely a technical access path; it’s a boundary at which an attacker's capabilities meet a system vulnerability and where a well-designed control can stop an attack before it reaches the components that matter most.

EP #	Entry Points	Attacker Profile	PHOENIX Component	How Attacker Exploits It	Attack Type	Risk	Dataset Evidence (Sprint 1 CTI)
EP-01	**External Sources of Cyber Threats & Scams**	Opportunistic Cybercriminals, Misinformation Actors	Data Ingestion Layer	Attacker floods the Scamwatch/cyber threat feed with bulk automated fake scam reports to manipulate the AI model's threat detection and produce false risk scores during a disaster.	Data Poisoning / Feed Manipulation	**High**	Row 9:900 bot applications to Red Cross $250M fund (2020). Row 25:3,600+ fraudulent flood recovery claims using computer-generated images.
EP-02	**External Sources of Disasters**	Organised Cybercrime, State-Sponsored APT	Data Ingestion Layer	Attacker intercepts the BoM/disaster feed and injects false hazard severity data before it reaches Data Validation, suppressing or falsely triggering Phoenix's threat elevation.	Tampering / Man-in-the-Middle	**High**	Row 54: Attackers manipulate IoT disaster sensors to provide false readings, causing incorrect disaster predictions (2023). Chakraborty et al. (2024).
EP-03	**DMT User Login**	Hacktivists, Insider Threats	Authentication System	Attacker uses credential stuffing, brute-force, or spear-phishing to compromise a DMT account and gain control of dashboard, alert approval, and risk threshold settings.	Credential Attack / Elevation of Privilege	**Critical**	Row 16: Stolen credentials used to access Australia's Early Warning Network, sending fake alerts to thousands (2019). Row 45: Unauthorised access via stolen credentials to send false emergency notifications (2024).
EP-04	**Community / Auths Feedback**	Misinformation Actors, Opportunistic Cybercriminals	Community Feedback Module	Attacker submits coordinated false feedback reports through the Community Feedback module — labelling genuine threats as false positives and false positives as genuine — to corrupt the AI model retraining pipeline over time.	Adversarial Feedback / Data Poisoning	**High**	Row 40: False bushfire narratives coordinated via social media causing public confusion (2019–20). Vicari & Komendatova (2023) — coordinated disinformation campaigns targeting AI systems.
EP-05	**Stream Processing (Kafka / Firebase / APIs)**	Organised Cybercrime, State-Sponsored APT	Stream Processing Layer	Attacker exploits misconfigured Kafka broker or Firebase security rules to inject malicious messages directly into the data stream after Data Validation has already passed, bypassing all ingestion controls.	API Abuse / Kafka Topic Poisoning	**High**	Row 20: Optus breach via unprotected API — 10 million Australians exposed (2022). Row 52: Unsecured APIs in disaster response platforms exploited to access victim records (2024).
EP-06	**Alert System (Send secure alert / API security & Encryption)**	State-Sponsored APT, Organised	TEAVS Alert Dispatch	Attacker intercepts, suppresses, or spoofs alert delivery API calls — stealing API credentials or exploiting	Alert Spoofing / Suppression / Credential Theft	**Critical**	Row 43: Fake emergency alert SMS impersonating government disaster warning system (2025).

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		Cybercrime, Hacktivists		alert endpoint vulnerabilities — to send fake verified alerts through official channels or block genuine ones.			Row 22: Fraudulent SMS impersonating Centrelink and Queensland Premier's official disaster relief appeal (2025). Chirgwin (2023) — FRV ransomware attack.
EP-07	**Scam DB / Cyber Threat DB**	Insider Threats, Organised Cybercrime	Backend Database Layer	Attacker exploits SQL injection, misconfigured access controls, or insider privileges to read, modify, or delete records in the Scam DB or Cyber Threat DB, directly corrupting the AI model's classification data.	SQL Injection / Unauthorised Access / Ransomware	**Critical**	Row 12: FRV ransomware attack encrypted systems across 85 stations — tax file numbers, bank details, health records leaked (2022). Row 26:3,000 NSW flood victims' personal data exposed via insecure handling (2025).
EP-08	**AI Model Processing (Threat Classification / Scam Detection / Sentiment / Urgency)**	State-Sponsored APT, Organised Cybercrime	ADCRS AI Engine	Attacker uses adversarial inputs or model evasion techniques — crafting scam content that exploits AI model weaknesses — to cause the model to consistently misclassify genuine threats as benign.	Adversarial ML / Model Evasion	**High**	Row 53:AI-generated deepfake videos spreading false disaster information — AI systems manipulated to produce misleading outputs (2024). Row 44:AI-generated phishing emails automating large-scale disaster scam campaigns (2025).
EP-09	**Role-based Access Control**	Insider Threats, State-Sponsored APT	Authorisation Layer	Attacker exploits overly permissive RBAC configuration or privilege escalation vulnerability to elevate a low-privilege account (Read-Only Analyst) to DMT Reviewer or System Administrator level.	Privilege Escalation / RBAC Misconfiguration	**Critical**	Row 12: FRV ransomware — attackers gained privileged access to critical emergency systems (2022). Spyros et al. (2025) — privilege escalation in CTI management systems.
EP-10	**Correlation Engine (Cross-refer scams & disasters / Risk Scoring)**	Misinformation Actors, State-Sponsored APT	Risk Correlation Engine	Attacker injects carefully crafted false inputs into the correlation engine to produce incorrect hazard-to-threat cross-references, generating a falsely low risk score during a real disaster to suppress alerts.	Logic Manipulation / Data Injection	**High**	Row 35: Hackers sent fake emergency alerts causing public panic and loss of trust in alert systems (2019). Row 54: Manipulated IoT sensors caused incorrect disaster predictions (2023).
EP-11	**Failure to Fetch Data from Sources (Retry arrows)**	Organised Cybercrime, Hacktivists	Error Handling / Retry Logic	Attacker deliberately triggers repeated data fetch failures, causing the retry mechanism to loop continuously and exhaust system resources — creating a denial of service during an active disaster.	Denial of Service / Resource Exhaustion	**Medium**	Row 33: DP World cyber-attack exhausted port systems — 40% of Australian freight disrupted (2023). Row 48: DDoS attacks targeting emergency response websites during crisis events (2024).

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EP-12	**Model Retraining (Feedback Collection loop)**	Insider Threats, Organised Cybercrime	Retraining Pipeline	Attacker submits a large volume of adversarially crafted feedback immediately before a scheduled model retraining cycle to maximise corruption of the retrained model before detection.	Feedback Timing Attack / Retraining Poisoning	**High**	Row 49: Data breach exposing disaster victim information from poorly secured databases used in downstream processing (2023). Azambuja et al. (2023) — adversarial feedback degrading AI model integrity.

Entry Point Analysis — Detailed Discussion

Entry Point 1 — External Cyber Threat & Scam Feed

Phoenix's external scam and cyber threat intelligence feed which drawing on Scamwatch, ASD threat feeds, and social media monitoring is the first and most exposed entry point in the system. Opportunistic cybercriminals and misinformation actors as identified in the attacker profiles are the primary threat actors at this boundary. Their strategy is not to attack Phoenix directly but to manipulate the information environment it monitors. During the 2019-20 Black Summer bushfires, bot accounts coordinated hundreds of fraudulent scam report submissions while simultaneously generating the social media misinformation that Phoenix is designed to detect. (Scamwatch, 2020). The 2022 floods saw over 3,600 fraudulent recovery grant claims submitted using computer-generated imagery and false documentation (Dataset Row 25). For Phoenix, the risk is that a sufficiently coordinated data poisoning campaign could train its AI model to recognise sophisticated scam patterns as legitimate, precisely when communities are most vulnerable. Mitigation requires source validation, submission rate limiting, and anomaly detection on sudden feed volume spikes.

Entry Point 2 — External Disaster Data Feed

The Bureau of Meteorology, CFA, Emergency Management Victoria and Geoscience Australia feeds provide the disaster context that gives Phoenix's threat correlations their meaning. As the cybersecurity department overview establishes, this component functions as a dynamic contextual trigger. The moment a significant disaster is detected, Phoenix elevates its monitoring sensitivity for that region. (Chakraborty et al., 2024). This makes the feed itself a high-value target. State-sponsored APT actors and organised cybercrime groups with the capability for man-in-the-middle attacks or DNS spoofing could intercept and manipulate disaster data in transit which causing Phoenix to either fail to elevate its threat response during a real disaster or to falsely elevate it during a non-event. Both outcomes that would erode community trust. Manipulated IoT disaster monitoring sensors have already demonstrated this attack vector in documented incidents (Dataset Row 54).

Entry Point 3 — DMT User Login

The DMT User Login is the gateway to Phoenix's most sensitive human-controlled functions like alert approval, model configuration, and data source management. The boundary is targeted by hacktivists seeking operational disruption, insider threats exploiting legitimate access and state-sponsored APT actors using spear-phishing and credential harvesting (Azambuja et al., 2023). The documented 2019 compromise of Australia's Early Warning Network where an attacker used stolen credentials to send fake emergency alerts to thousands of Queensland residents which is the most directly relevant precedent for this entry point (Dataset Row 16).The Phoenix cybersecurity architecture responds to this with multi-factor authentication and role-based access control which the entry point analysis confirms these controls are correctly directed at the highest-risk access boundary in the system.

Entry Point 4 — Community Feedback Module

The Community and Auth Feedback module is the entry point that misinformation actors and opportunistic cybercriminals are most likely to target indirectly. Because this channel feeds directly into Phoenix's model retraining pipeline, a coordinated adversarial feedback campaign by submitting false threat confirmations and false positive labels simultaneously can gradually shift the AI model's classifications without ever directly attacking the system's infrastructure. Vicari and Komendatova's (2023) meta-analysis confirms that coordinated disinformation campaigns targeting AI systems through feedback channels are a documented and growing attack vector. The 2019-20 bushfire misinformation campaign, which saw false narratives coordinated across social media with bot amplification, demonstrates the scale at which such campaigns can be executed in an Australian disaster context (Dataset Row 40).

Entry Point 5 — Stream Processing (Kafka / Firebase / APIs)

The stream processing layer, Apache Kafka for high-throughput event routing and Firebase for community-facing interfaces sits downstream of Data Validation and is therefore a target for attackers seeking to inject data into the pipeline after it has already passed security checks. Kafka topic poisoning and Firebase unauthorised access are among the attack types. The Optus data breach which caused by an unprotected API endpoint that bypassed standard access controls demonstrates how API misconfiguration can expose a system's internal data pipelines to external exploitation at scale (Dataset Row 20). For Phoenix, a compromised stream processing layer means attacker-controlled data can reach the AI model and correlation engine without triggering any validation alarms.

Entry Point 6 — Alert System

The Alert System is the most consequential entry point in Phoenix which the system's analysis becomes public action. Alerts are delivered through encrypted channels including SMS, email, push notifications via Firebase, and integration with VicEmergency, all authenticated using OAuth 2.0 and TLS 1.3 (Chirgwin, 2023; Nguyen & Bogle, 2020). The December 2022 ransomware attack on Fire Rescue Victoria demonstrated precisely what is at stake when this boundary is breached and communications disrupted, dispatch systems offline, and 85 stations forced into manual operation for months (Dataset Row 12). Alert suppression, spoofing, and content tampering are all catalogued as Critical severity. The 2025 incidents of fraudulent SMS messages impersonating Centrelink and the Queensland Premier's office confirm that alert channel spoofing is not theoretical, it’s happening in Australia right now (Dataset Row 22).

Entry Point 7 — Scam DB and Cyber Threat DB

The Scam DB and Cyber Threat DB are the repositories that ground Phoenix's AI model in historical threat intelligence and confirmed scam patterns. Insider threats and organised cybercrime groups both identified in as targeting Trust Boundary 2 which represent the primary risks at this entry point. SQL injection, unauthorised direct write access, and ransomware encryption are all catalogued attacks relevant here. The 2025 exposure of NSW flood victims' personal data which uploaded to an external platform by a contractor illustrates that database boundary violations do not always require sophisticated technical attacks; they can arise from negligent insider behaviour that bypasses every technical control (Dataset Row 26). Ransomware targeting emergency services databases in the Fire Rescue Victoria incident confirms the operational impact when this boundary fails.

Entry Point 8 — AI Model Processing

The AI Model Processing component which running threat classification, scam detection, sentiment analysis, and urgency detection is the analytical core of Phoenix and the entry point most vulnerable to adversarial machine learning attacks. State-sponsored APT actors and organised cybercrime groups with sufficient technical capability can probe the model's decision boundaries through black-box testing and generate adversarial content that consistently evades detection. Attack list catalogues model inversion, membership inference, and confidence score manipulation as specific attack vectors here. The emergence of AI-generated phishing emails by using large language models to generate convincing, contextually tailored scam messages at scale is directly documented in the Sprint 1 dataset (Dataset Row 44) and represents an evolving adversarial capability that Phoenix's scam detection model must continuously adapt to counter.

Entry Point 9 — Role-Based Access Control

Role-Based Access Control governs what authenticated DMT users can see and do by separating Read-Only Analysts, DMT Reviewers, and System Administrators into distinct permission tiers. A misconfigured or overly permissive RBAC implementation is the single most dangerous internal vulnerability in Phoenix, because it can render every other security control irrelevant. An insider threat who successfully escalates from Read-Only Analyst to System Administrator can suppress alerts, alter model configurations, or exfiltrate threat intelligence without triggering any external security alarms. Privilege escalation within RBAC systems is catalogued as Critical severity and the principle of least privilege granting only the minimum access required for each role is the primary technical countermeasure.

Entry Point 10 — Correlation Engine

The Correlation Engine is where Phoenix's most consequential analytical work takes place by cross-referencing active threat detections against live disaster event data to classify event-correlated threats and generate risk scores. Because the Correlation Engine's outputs directly determine which threats escalate to DMT review and which are automatically routed or discarded, manipulating its inputs produces outsized downstream effects. An attacker who can suppress Phoenix's correlation of a genuine disaster-cyber threat correlation. For example, by injecting false timestamp or geolocation data into the feeds can prevent a critical alert from ever reaching human review. The 2019 fake emergency alert incident demonstrates the real-world consequence of compromised threat correlation in an emergency management context (Dataset Row 35).

Entry Point 11 — Failure to Fetch Data / Retry Mechanism

The automatic retry mechanism triggered when data cannot be fetched due to network disruption, API downtime, or denial-of-service conditions is a necessary resilience feature that simultaneously creates a denial-of-service vulnerability. An attacker who can deliberately trigger repeated fetch failures forces Phoenix into a continuous retry loop that exhausts system resources and creates monitoring gaps at exactly the moments those gaps are most exploitable. The 2023 DP World cyber-attack, which forced shutdown of Australian port operations by overwhelming system resources demonstrates the operational impact when retry and failover mechanisms are weaponised against critical infrastructure (Dataset Row 33). The Phoenix cybersecurity architecture mitigates this through exponential backoff and maximum retry limits.

Entry Point 12 — Model Retraining

The Model Retraining pipeline which driven by DMT review decisions, community feedback, and automated system logs is how Phoenix stays ahead of an adversary landscape that never stands still. It's also the entry point most vulnerable to slow, patient, long-term attacks that may not be detectable until significant model degradation has already occurred. A feedback timing attack by submitting coordinated adversarial feedback immediately before a scheduled retraining cycle can corrupt the retrained model before the corrupted feedback is identified. Misinformation actors and insider threats targeting the retraining pipeline, and the Discard if Irrelevant component exists precisely because feedback manipulation is a recognised attack vector (Azambuja et al., 2023; Spyros et al., 2025).

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